Spacecraft Controls and Root Locus Analysis

Potratz, J., Sharp, R., and Speredelozzi, N.

Spacecraft Controls Fall 2025. Embry-Riddle Aeronautical University, Daytona Beach, Florida.

November 25, 2025

This project uses root locus methods presented in Ogata's *Modern Control Engineering [1]*, to demonstrate how the closed-loop poles of open-loop systems move as feedback gain varies. Using constants $\boldsymbol{p = 9, q = 3}$, and $\boldsymbol{r = 1}$, we defined three systems that share open-loop pole locations at –3 and –1 on the real axis. Additionally, the first and third systems include open-loop zeroes at –9 and –0.5 respectively. Our task one method included hand calculations for positive gain value $\boldsymbol{K}$ that yielded the minimum damping ratio $\boldsymbol{\zeta_{min} = 0.638}$, as well as minimum *2%* settling time $\boldsymbol{t_{s_{min}}(2\%) = 1.091\ s}$. Analysis for task two focused on root-locus behavior and characteristics such as asymptote angles, centroid of asymptotes on the real axis, breakaway points, and imaginary-axis crossings. The hand calculated results were then verified using MATLAB. For the second system, three asymptotes were found in addition to a real-axis breakaway point between the middle poles. Imaginary-axis crossings were also present, suggesting the closed-loop response becomes unstable for certain gain values. With the third system, the addition of an open-loop zero produced only two asymptotes, shifted the root loci, and removed the imaginary-axis crossing, suggesting that the system would be stable for any positive gain value. The root loci generated by MATLAB agreed with the hand-drawn results for both systems.

Introduction

Control systems are imperative for spacecraft response to commands and external disturbances. Understanding how system behavior changes with feedback is an important part of classical control theory [1]. One useful tool for the analysis of system response is the root-locus method, which shows how a system's poles move on the s-plane as the feedback gain varies. The s-plane is a coordinate system in which the horizontal line indicates stability in a system; points farther left lead to more stable systems, and the vertical line represents oscillatory behavior. By examining how poles move across this plane, predictions can be made regarding the change in stability and oscillatory behavior of the system as gain varies.

In this project, we use constants $\boldsymbol{p = 9, q = 3}$, **and** $\boldsymbol{r = 1}$, determined from our group member's birth months to define three related systems. The first system used in Task 1 is $G(s) = \frac{s+p}{(s+q)(s+r)}$.

Task 1 focuses on closed-loop pole analysis. For this system we found the characteristic equation with a positive feedback gain, (K) and examined how the poles depend on K. The main goals of this task were to find the gain that produces the minimum damping ratio (ζ), to determine the corresponding 2% settling time and to investigate how aggressive the closed-loop response could get before the system becomes underdamped. The results were calculated using second order approximations [1] and verified using MATLAB. The remaining two systems denoted below were evaluated in tasks 2a) and 2b) respectively.

$$G(s) = \frac{1}{(s+p)(s+q)(s+r)} \text{ (Task 2(a)) and } G(s) = \frac{s+\frac{r}{2}}{(s+p)(s+q)(s+r)} \text{ (Task 2(b))}$$

These transfer functions represent two open-loop systems that share the same three poles but differ by the addition of an open-loop zero in Task 2(b). For these systems, we applied the root locus method to see how the closed-loop poles are affected as the gain K increases. We focused on identifying the important geometric features of the root locus, such as the asymptote angles, the centroid location of asymptotes on the real axis, breakaway points, and any imaginary-axis crossings. The hand drawn root loci seen in appendix B were compared to results produced by MATLAB's "rlocus" function to validate the analysis.

Overall, the purpose of this project is to connect the mathematical description of each system to its graphical behavior on the s-plane. Task 1 shows how gain affects damping and settling time in a closed-loop system, while tasks 2a) and 2b) show how pole and zero locations shape the root locus and in turn, the oscillatory behavior and stability of the closed-loop response. These tasks provide a clear representation of how classical root locus methods can be used to examine and improve spacecraft control systems.

Theoretical Development

<u>Part 1:</u>

In this project, we use constants $\boldsymbol{p = 9, q = 3}$, **and** $\boldsymbol{r = 1}$, determined from our group member's birth months to define three related systems. The first system used in Task 1 is shown in Equation (1).

$$G(s) = \frac{s+p}{(s+q)(s+r)} \qquad \text{Equation (1)}$$

Inserting the constants into Equation (1), we get Equation (2).

$$G(s) = \frac{s+9}{(s+3)(s+1)} \qquad \text{Equation (2)}$$

To find the value of K that will yield the minimum damping ratio, the positive gain K is inserted into Equation (2), which results in Equation (3).

$$G(s) = \frac{K(s+9)}{(s+3)(s+1)} \qquad \text{Equation (3)}$$

The characteristic equation can then be found by adding the numerator and denominator and setting them equal to 0 as shown in Equation (4).

$$K(s+9) + (s+3)(s+1) = 0 \qquad \text{Equation (4)}$$

Manipulating Equation (4) to match the standard form shown in Equation (5), coefficient matching can be used to determine the value of ζ, which is the damping ratio.

$$s^2 + (2\zeta\omega_n)s + \omega_n^2 = 0 \qquad \text{Equation (5)}$$

$$s^2 + (4+K)s + 3 + 9K = 0 \qquad \text{Equation (6)}$$

Equation (6) shows the manipulated equation to match the standard form. Utilizing coefficient matching, you can then solve for ω_n, the undamped natural frequency and ζ in terms of K. Solving for both yields the following equations, with ω_n shown in Equation (7) and ζ shown in Equation (8).

$$\omega_n = \sqrt{3+9K} \qquad \text{Equation (7)}$$

$$\zeta = \frac{(4+K)}{2\sqrt{3+9K}} \qquad \text{Equation (8)}$$

By squaring both sides of Equation (8), we determine a function that is called $f(K)$. This is shown in Equation (9).

$$f(K) = \zeta^2 = \frac{(4+K)}{4(3+9K)} \quad \text{Equation (9)}$$

By simplifying and taking the derivative of $f(K)$ and setting it equal to 0, we can solve for the value of K that will result in the minimum damping ratio ζ . The equation that results from this derivative is shown as Equation (10).

$$36K^2 + 24K - 480 = 0 \qquad \text{Equation (10)}$$

Using the quadratic formula, the discriminant is determined to be positive, meaning that there will be 2 real solutions. After applying the quadratic equation to solve for K, one real solution is found to be negative, and the other positive. The negative value can be ignored, since a positive gain value means that we need a positive K value. The two solutions are shown below in Equation (11).

$$K = \frac{240}{72},\ K = -\frac{288}{72} \quad \text{Equation (11)}$$

After determining the correct value for K, it is used to determine the value of ζ_{min}, which is the minimum damping ratio. This value of K is also used to determine the undamped natural frequency, ω_n, which is then used to determine the minimum 2% settling time by using Equation (12).

$$t_{s_{min}}(2\%) = \frac{4}{\zeta_{min}\omega_n} \quad \text{Equation (12)}$$

<u>Part 2a:</u>

For task 2 part a, a new equation is used.

$$G(s) = \frac{1}{(s+p)(s+q)(s+r)} \text{Equation (13)}$$

Inserting our unique constants and inserting the positive gain K into Equation (13), it results in Equation (14).

$$G(s) = \frac{K}{(s+9)(s+3)(s+1)} \text{Equation (14)}$$

Equation (15) is then used to solve for the open loop poles of the system.

$$(s+9)(s+3)(s+1) = 0 \quad \text{Equation (15)}$$

Solving Equation (15) for s results in three values; $s_1 = -1,\ s_2 = -3,\ s_3 = -9$. Note that all 3 open-loop poles are on the real axis, and that there are 0 open loop zeroes. Defining n as the number of poles and m as the number of zeroes, the center of asymptotes on the real axis,$\sigma_\circ$, can be solved for using Equation (16).

$$\sigma_\circ = \frac{(\Sigma_{poles} - \Sigma_{zeroes})}{n-m} = \frac{(-1-3-9-0)}{3-0} = -\frac{13}{3} \qquad \text{Equation (16)}$$

Next, the angles of the asymptotes were calculated using Equation (17).

$$\theta_{asymptote} = \pm\frac{180(2K+1)}{n-m} \qquad \text{Equation (17)}$$

$\theta_{asymptote}$ was then calculated for different values of K, starting at 0, then 1, then 2 and so on. However, repeating asymptote angles were ignored, meaning that the only 3 asymptote angles were $\theta_1 = 60°$, $\theta_2 = -60°$, and $\theta_3 = 180°$. Imaginary crossings were then found using Equation (18) and setting it equal to 0.

$$P(s) = s^3 + 13s^2 + 39s + 27 + K = 0 \qquad \text{Equation (18)}$$

Using Routh's stability criterion to solve for K, s is found to be $\pm 6.245j$. These are intercepts for the imaginary crossings. The breakaway points were then found by taking the derivative of K in Equation (19) and setting it equal to 0.

$$K = -\frac{1}{G(s)} = -(s+9)(s+3)(s+1) \qquad \text{Equation (19)}$$

Solving this derivative, two values of s are determined. Only the real breakaway point is taken, which is when s < -9 or –3 < s < -1. The root locus was then drawn on a graph by hand and graphed in MATLAB.

<u>Part 2b:</u>

The function used for part 2b is Equation (20).

$$G(s) = \frac{s+\frac{r}{2}}{(s+p)(s+q)(s+r)} \text{Equation (20)}$$

After inserting the constants and the positive gain K we get Equation (21).

$$G(s) = \frac{K\left(s+\frac{1}{2}\right)}{(s+9)(s+3)(s+1)} \text{Equation (21)}$$

The characteristic equation can then be found and used to solve for the open loop poles. The characteristic equation is shown as Equation (22).

$$(s+9)(s+3)(s+1) = 0 \quad \text{Equation (22)}$$

The open loop poles are found at $s_1 = -1$, $s_2 = -3$, and $s_3 = -9$. The number of open loop poles is denoted as 'n' meaning that in this case $n = 3$. The open loop zeroes can also be found by solving for s using Equation (23).

$$\left(s + \frac{1}{2}\right) = 0 \quad \text{Equation (23)}$$

Solving Equation (23) yields $s = -\frac{1}{2}$. Now that the open loop poles and zeroes have been determined, the center of asymptotes on the real axis, can be solved for using Equation (16).

$$\sigma_\circ = \frac{(\Sigma_{poles} - \Sigma_{zeroes})}{n-m} = \frac{\left(-1-3-9-\left(-\frac{1}{2}\right)\right)}{3-1} = -\frac{12.5}{2} \qquad \text{Equation (16)}$$

The angles of asymptotes can also be determined using Equation (17).

$$\theta_{asymptote} = \pm\frac{180(2K+1)}{n-m} \qquad \text{Equation (17)}$$

K was valued as 0, then 1, then 2 and so on. However, repeating angles were ignored. In this case, two distinct asymptotes were found, $\theta_1 = 90°$and $\theta_2 = -90°$. Next, imaginary crossings were found using the characteristic polynomial $P(s)$, which is defined in Equation (24).

$$P(s) = s^3 + 13s^2 + 39s + 27 + Ks + \frac{K}{2} = 0 \qquad \text{Equation (24)}$$

Using Routh's stability criterion, K was determined to be equal to -38.4. This negative K value determines that there are no intercepts with the imaginary axis. The breakaway points can then be determined using Equation (25).

$$K = -\frac{1}{G(s)} = -\frac{(s+9)(s+3)(s+1)}{\left(s+\frac{1}{2}\right)} \qquad \text{Equation (25)}$$

Taking the derivative of Equation (25) and setting it equal to 0 yields three s values. Only the s values in which –3 < s < -9, or –1.5 < s < -0.5 were real breakaway points in which the locus existed. These were then drawn on a graph by hand and also in MATLAB.

Results

Task 1 utilizes closed-loop pole analysis of the open-loop transfer function denoted in Equation (3) to solve for the positive gain value $K_{min} = 3.333$ that yielded a minimum damping ratio of $\zeta_{min} = 0.638$. The corresponding undamped natural frequency $\omega_n = 5.744\frac{rad}{s}$ was used to determine the minimum 2% settling time of the system $t_{s_{min}}(2\%) = 1.091\ s$. The closed-loop poles associated with this optimal gain value were located at $s = -3.665 \pm 4.423j$, indicating a stable, underdamped system.

Tasks 2a and 2b focused on root locus behavior of the open-loop transfer functions defined in Equations (14) and (21) respectively. Appendix B contains hand-drawn root loci for each system and all necessary calculations to determine the angle of each asymptote, the centroid of the asymptotes on the real axis, any break-in/break-away points, and any imaginary axis crossings. The root locus for task 2a's open-loop transfer function was graphed in MATLAB using the built-in "rlocus" function as shown in figure 1 below.

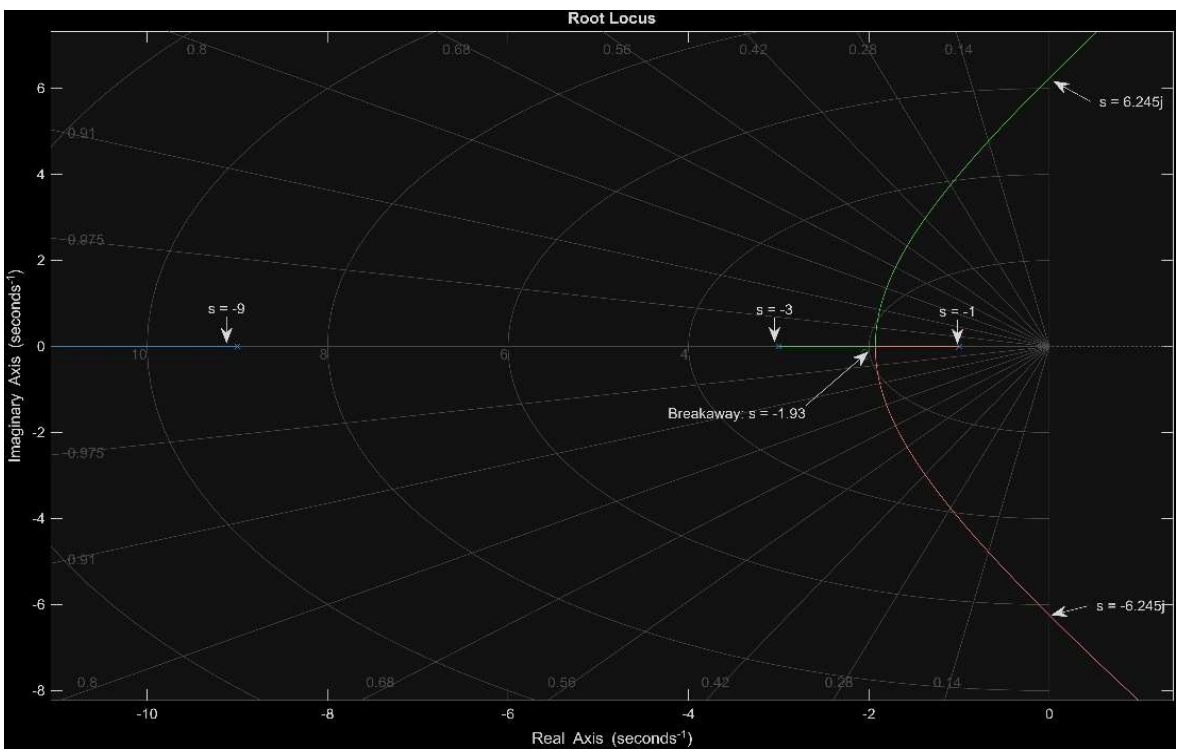


Figure 1: Task 2a root locus graphed in MATLAB

Three open-loop poles were identified at $s = -9, -3$, and -1 with no open-loop zeros. The root locus also exhibited three asymptotes with angles of $60°, -60°$, and $180°$ and a centroid on the real axis at $\sigma_0 = -4.333$. A singular breakaway point was located at $s = -1.93$ between the two rightmost poles. Imaginary axis crossings were also calculated using Routh's stability criterion, intersecting at $s = \pm 6.245j$. Their presence indicates instability in the system for large gain values. Figure 2 displays the task 2b root locus.

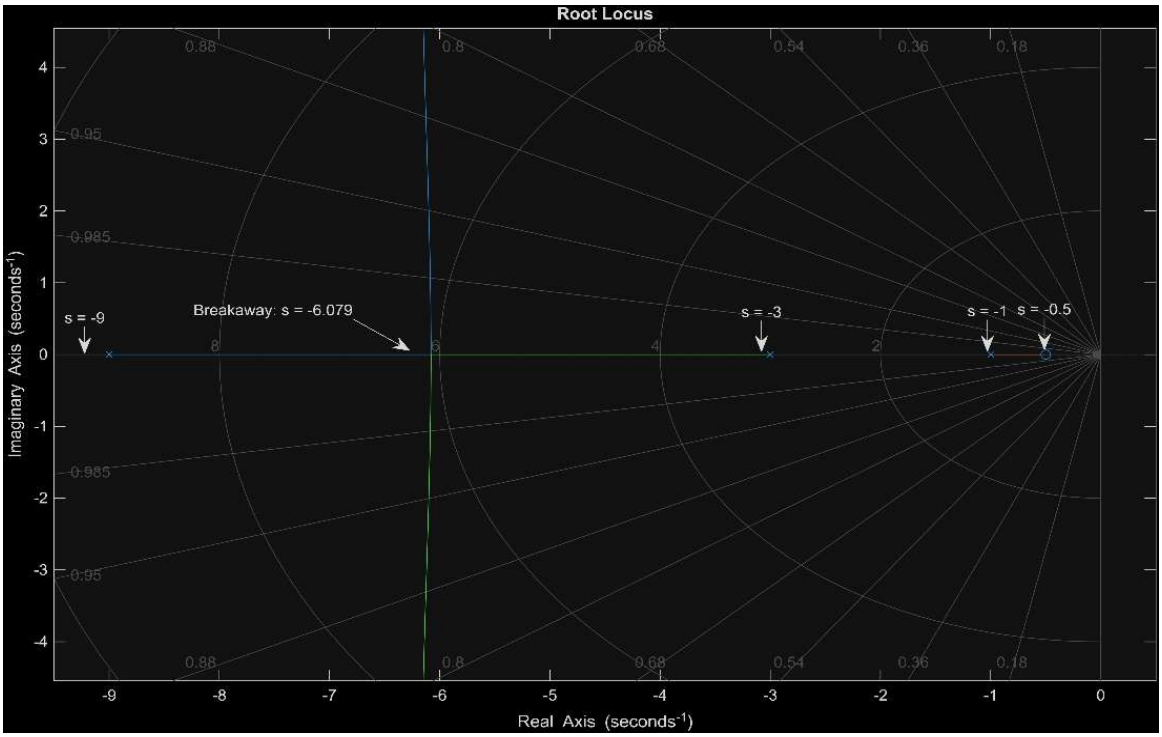


Figure 2: Task 2b root locus graphed in MATLAB

The root locus exhibited by Equation (21) yielded the same open-loop poles at $s = -9, -3,$ and -1, with the addition of an open-loop zero at $s = -0.5$. This change eliminates an asymptote and changes the remaining two asymptote angles to $\pm 90°$. The real-axis centroid location was also shifted to $\sigma_0 = -6.25$, with a breakaway point nearby at $s = -6.079$. Contrary to task 2a, Routh's stability criterion yields no imaginary axis crossings, indicating stability in the system for all positive gain values.

Conclusions

This project successfully demonstrated how classical control theory and root-locus methods are used to model closed-loop dynamic behavior of various systems. In task 1, the relationship between positive gain K, damping ratio ζ, and 2% settling time $t_s(2\%)$ was established through second-order approximation. The optimal gain value $K_{min} = 3.333$ achieved a balance between response speed and oscillatory behavior, yielding a fast settling time of approximately 1.091 seconds without excessive overshoot.

Tasks 2a and 2b examined the behavior of root loci and their strong dependance on the presence of open-loop zeros. The system without zeros exhibited three asymptotes and crossed the imaginary axis, leading to instability for large gain values. Whereas the introduction of a single zero in task 2b reduced the number of asymptotes, shifted the locus negatively along the real axis, and eliminated all imaginary axis crossings, resulting in stable system behavior for all positive gain values.

Overall, the project illustrates how root locus analysis offers both qualitative insight and quantitative design results for feedback control systems. The results confirm that careful pole-zero placement and gain selection are critical tools in shaping transient response behavior and ensuring stability in spacecraft control applications. The agreement between calculated predictions and MATLAB simulations demonstrates the reliability of classical control theory for real-world aerospace systems.

References

[1] K. Ogata, *Modern Control Engineering* 3rd Edition, Prentice Hall, (2010).

Appendices

Appendix A: MATLAB code for root locus graphs

```
%{
Group Members:
Section 07DB: Nick Speredelozzi, Riley Sharp
Section 06DB: Joshua Potratz

AE 434 Final Project
November 6th, 2025
%}

%% Derived Constants
clc
clear
close all

a = 9;                              % September birth month
b = 1;                              % January birth month
c = 3;                              % March birth month

p = max([a b c]);                   % Derived constant #1
r = min([a b c]);                   % Derived constant #2
q = c;                              % Derived constant #3

%% Task 2a.) Consider a system with the open loop transfer function G(s):
% Part i: Completed graphically by hand
% Part ii: Graph the root locus in MATLAB using the built-in function

s = tf('s');                        % Define the Laplace variable for G(s)
G = 1 / ((s+9)*(s+3)*(s+1));        % Define the open loop transfer function G(s)
rlocus(G);                          % Plot the root locus of the transfer function

%% Task 2b.) Consider a system with the open loop transfer function G(s):
% Part i: Completed graphically by hand (indicated angle of each asymptote, center of
asymptotes on the real axis, break-in/break-away points, and imaginary crossings.
% Part ii: Graph the root locus in MATLAB using the built-in function

s2 = tf('s');                                       % Define the Laplace variable for G(s)
G2 = (s2 + 0.5*r) / ((s2+9)*(s2+3)*(s2+1));         % Define the open loop transfer function
G(s)
rlocus(G2);                                         % Plot the root locus of the transfer
function
```

Appendix B: Hand calculations/root locus behavior for tasks 1 and 2.

1. **Closed-Loop Pole Analysis:**

Consider a system with the open-loop transfer function:

$$G(s) = \frac{s+p}{(s+q)(s+r)} \tag{1}$$

a) Out of all the possible closed-loop pole locations for positive gain K, solve the value of K that will yield the minimum damping ratio.

b) Solve for the minimum 2% settling time.

1 a.) Derived constants: $p = 9,\ q = 3,\ r = 1$

$$G(s) = \frac{(s+p)}{(s+q)(s+r)} = \frac{(s+9)}{(s+3)(s+1)} = \frac{K(s+9)}{(s+3)(s+1)}$$

Characteristic equation:

$$\frac{K(s+9)}{(s+3)(s+1)} \longrightarrow (s+3)(s+1) + K(s+9) = 0$$

$$0 = s^2 + 4s + 3 + Ks + 9K = s^2 + (4+K)s + 3 + 9K$$

Standard form: $s^2 + (2\zeta\omega_n)s + \omega_n^2 = 0$

Coefficient matching: Goals: $(+K) \rightarrow$ positive gain

minimal ζ (damping ratio)

$$\omega_n^2 = 3 + 9K$$

$$\omega_n = \sqrt{3+9K}$$

$$(2\zeta\omega_n)\cancel{s} = (4+K)\cancel{s} \longrightarrow \zeta = \frac{4+K}{2\sqrt{3+9K}}$$

$$\zeta^2 = \frac{(4+K)^2}{4(3+9K)} \longrightarrow \text{Let } f(K) = \frac{(4+K)^2}{(3+9K)}$$

For minimum ζ, determine $f'(K) = 0$

$$\frac{d\zeta^2}{dK} = \frac{v\,du - u\,dv}{v^2}$$

$u = (4+K)^2 \quad du = 2(4+K)(1) = (8+2K)$

$v = (12+36K) \quad dv = 36$

$$f'(K) = \frac{(12+36K)(8+2K) - 36(4+K)^2}{(12+36K)^2} = 0$$

$$(12+36K)(8+2K) - 36(K^2+8K+16) = 0$$

$$0 = 96 + 24K + \cancel{288K} + 72K^2 - 36K^2 - \cancel{288K} - 576$$

$$0 = 36K^2 + 24K - 480$$

<u>Quadratic Formula</u>: $K = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$K = \frac{-24 \pm \sqrt{(24)^2 - 4(36)(-480)}}{2(36)} = \frac{-24 \pm \sqrt{69,696}}{72}$$

discriminant is (+), so K has 2 real solns.

$$K = \frac{-24 \pm 264}{72} = \frac{240}{72}, \cancel{-\frac{288}{72}}$$

positive gain means (+K)

$$K = \frac{10}{3} = 3.333$$

$\boxed{K_{min} = 3.333}$ part a.) positive gain value yielding the minimum damping ratio (ζ)

$$\zeta_{min} = \frac{4 + K_{min}}{2\sqrt{3 + 9K_{min}}} = \frac{4 + 3.333}{2\sqrt{3 + 9(3.333)}} \boxed{= 0.638}$$ minimum damping ratio

<u>Settling time (2%) criterion</u>: $t_{s_{min}}(2\%) = \dfrac{4}{\zeta_{min}\,\omega_n}$

$$\omega_n = \sqrt{3 + 9K} = \sqrt{3 + 9(3.333)} \boxed{= 5.744 \ \frac{rad}{s}}$$ undamped natural frequency

$$t_{s_{min}}(2\%) = \frac{4}{(0.638)(5.744)} \boxed{= 1.091 \text{ seconds}}$$ settling time (2%)

2. **Root Locus:**

a) Consider a system with the open-loop transfer function:

$$G(s) = \frac{1}{(s+p)(s+q)(s+r)} \tag{2}$$

i Draw the root locus for positive gain, indicating the angle of each asymptote, center of the asymptotes on the real axis, break-in/break-away points (if any), and imaginary crossings (if any).

ii Graph the root locus in MATLAB using the built- in function.

2a.i). Derived constants: $p = 9$, $q = 3$, $r = 1$

$$G(s) = \frac{1}{(s+9)(s+3)(s+1)} = \frac{K}{(s+9)(s+3)(s+1)}$$

Characteristic equation: $(s+9)(s+3)(s+1) = 0$

Positive Gain: ($\uparrow^{+}K$)

Open loop poles: 3 Open loop zeros: 0

$s_1 = -1$, $s_2 = -3$, $s_3 = -9$ (all on real axis)

$n = 3$ (# of poles/loci)

$m = 0$ (# of zeros)

$n - m = 3$ (# of distinct asymptotes)

Center of asymptotes on real axis:

$$\sigma_0 = \frac{\Sigma \text{poles} - \Sigma \text{zeros}}{n - m}$$

$$\sigma_0 = \frac{((-1) + (-3) + (-9)) - 0}{3}$$

$$\boxed{\sigma_0 = \frac{-13}{3} \approx -4.333}$$

Angles of asymptotes:

$$\theta_{asymptote} = \frac{\pm 180^\circ(2k+1)}{n-m}, \quad (k = 0, 1, 2 \ldots)$$

$$\theta_{asymptote}\Big|_{k=0} = \pm \frac{180^\circ(2(0)+1)}{3} = \pm 60^\circ \qquad \theta_{asymptote}\Big|_{k=3} = \pm \frac{180^\circ(2(3)+1)}{3} = \pm 420^\circ$$

(also equal to $\pm 60^\circ$)

$$\theta_{asymptote}\Big|_{k=1} = \pm \frac{180^\circ(2(1)+1)}{3} = \pm 180^\circ$$

$$\theta_{asymptote}\Big|_{k=2} = \pm \frac{180^\circ(2(2)+1)}{3} = \pm 300^\circ \text{ (equal to } \pm 60^\circ)$$

☆ angle repeats infinitely, so the 3 asymptotes have angles of:

$$\boxed{\theta_1 = 60^\circ,\ \theta_2 = -60^\circ,\ \theta_3 = 180^\circ \text{ (- real axis)}}$$

Imaginary Crossings: Characteristic equation: $P(s)$

$$P(s) = s^3 + 13s^2 + 39s + 27 + K = 0$$

Routh's Stability Criterion: $\frac{480 - K}{13} \longrightarrow K = 480$

$$13s^2 + (27 + K) = 0 \longrightarrow \boxed{s = \pm 6.245j} \text{ (imaginary axis intercepts)}$$

Breakaway points: $K = -\frac{1}{G(s)} = -(s+9)(s+3)(s+1)$

$$K = -(s^3 + 13s^2 + 39s + 27)$$

$$\frac{dK}{ds} = -(3s^2 + 26s + 39)$$

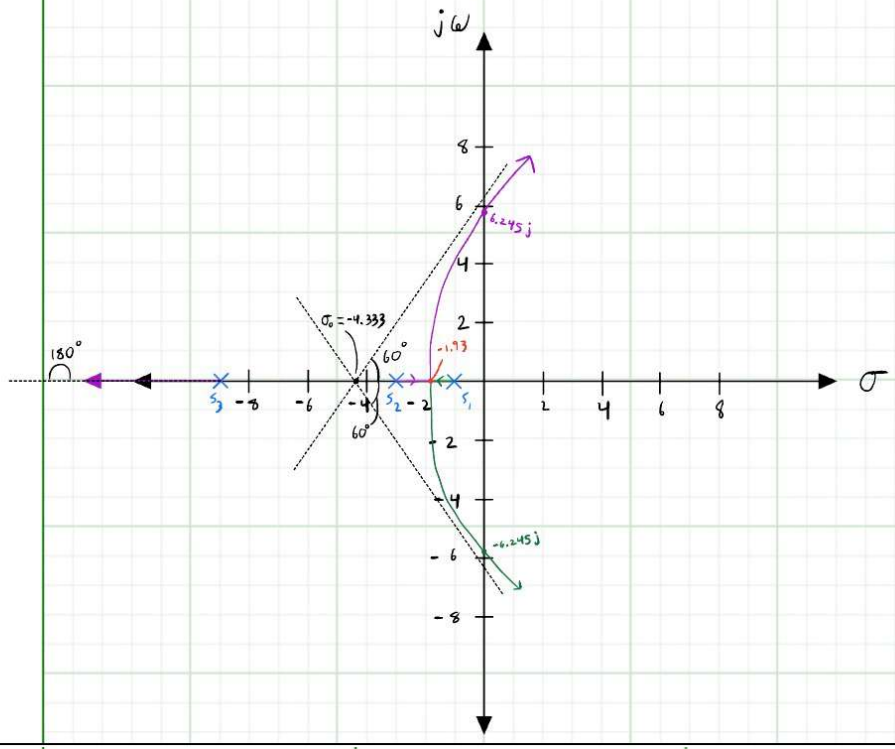
$$\frac{dK}{ds} = 0 = 3s^2 + 26s + 39 \xrightarrow[\text{formula}]{\text{quadratic}} s = -1.93 \quad s = -6.74$$

real breakaway ✓ breakaway doesn't exist ✗

S-Plane:

Locus: Exists: $s < -9$, $-3 < s < -1$

Doesn't exist: $-3 < s < -9$, $s > -1$



b) Consider a system with the open-loop transfer function:

$$G(s) = \frac{s + \frac{r}{2}}{(s+p)(s+q)(s+r)} \tag{3}$$

i Draw the root locus for positive gain, indicating the angle of each asymptote, center of the asymptotes on the real axis, break-in/break-away points (if any), and imaginary crossings (if any).

ii Graph the root locus in MATLAB using the built-in function.

Derived constants: $p = 9,\ q = 3,\ r = 1$

$$G(s) = \frac{(s + \frac{1}{2})}{(s+9)(s+3)(s+1)} = \frac{K(s + \frac{1}{2})}{(s+9)(s+3)(s+1)}$$

Characteristic equation:

$(s+9)(s+3)(s+1) = 0$

Positive Gain: $(\uparrow^{+} K)$

Open loop poles: 3

$s_1 = -1,\ s_2 = -3,\ s_3 = -9$ (all on real axis)

Open loop zeros: $(s + \frac{1}{2}) = 0$

$s = -0.5$

$n = 3$ (# of poles/loci)

$m = 1$ (# of zeros)

$n - m = 2$ (# of distinct asymptotes)

Center of asymptotes on real axis:

$$\sigma_0 = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$$

$$\sigma_0 = \frac{((-1) + (-3) + (-9)) - (-\frac{1}{2})}{2}$$

$$\boxed{\sigma_0 = -\frac{12.5}{2} \approx -6.25}$$

Angles of asymptotes:

$$\theta_{asymptote} = \frac{\pm 180^\circ(2k+1)}{n-m},\ (k = 0, 1, 2 \ldots)$$

$$\theta_{asymptote}\Big|_{k=0} = \pm \frac{180^\circ(2(0)+1)}{2} = \pm 90^\circ$$

$$\theta_{asymptote}\Big|_{k=1} = \pm \frac{180^\circ(2(1)+1)}{2} = \pm 270^\circ \text{ (equal to } \pm 90^\circ)$$

$$\theta_{asymptote}\Big|_{k=2} = \pm \frac{180^\circ(2(2)+1)}{2} = \pm 450^\circ \text{ (equal to } \pm 90^\circ)$$

Angle repeats infinitely, so the 2 distinct asymptotes have angles:

$$\boxed{\theta_1 = 90^\circ,\ \theta_2 = -90^\circ}$$

Imaginary Crossings: Characteristic polynomial: $P(s)$

$$P(s) = s^3 + 13s^2 + 39s + 27 + Ks + \frac{K}{2} = 0$$

Routh's Stability Criterion:

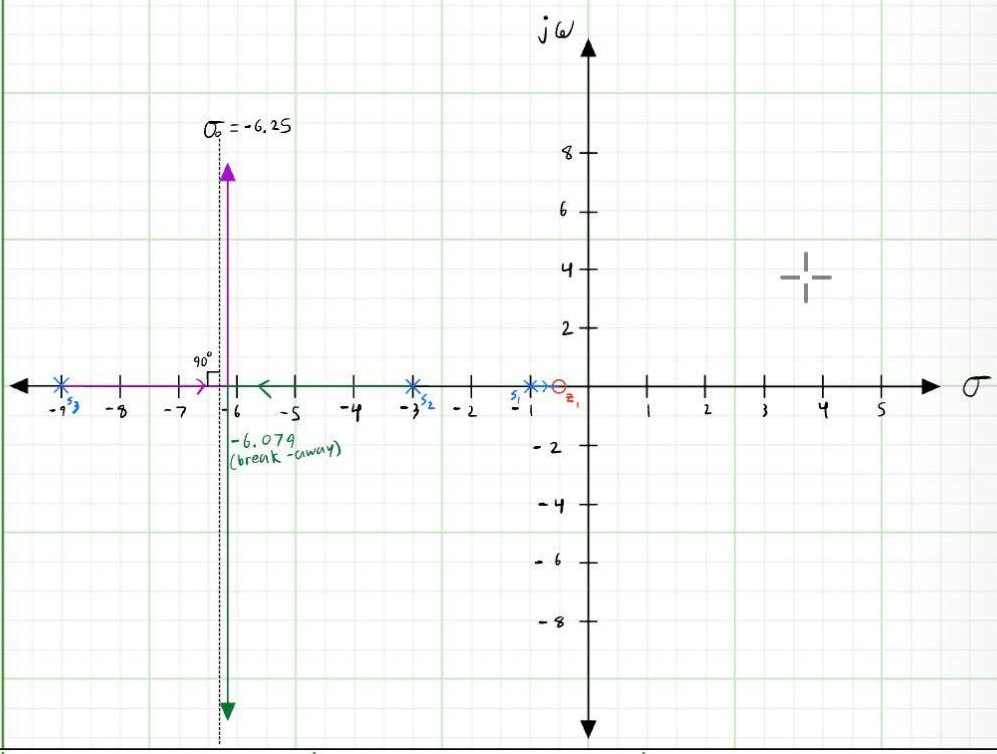
$480 + 12.5K = 0 \longrightarrow K = -38.4$ (negative K indicates no intercepts with imaginary axis)

Breakaway points: $K = -\frac{1}{G(s)} = \frac{-(s+9)(s+3)(s+1)}{(s+\frac{1}{2})}$

$$\frac{dK}{ds} = 0 = (3s^2 + 26s + 39)(s + \tfrac{1}{2}) - (s^3 + 13s^2 + 39s + 27) \qquad K = -\frac{(s^3 + 13s^2 + 39s + 27)}{(s+\frac{1}{2})}$$

$$\frac{dK}{ds} = 0 = (4s^3 + 29s^2 + 26s - 15)$$

$\xrightarrow[\text{Solver}]{\text{Cubic}}$ nonreal breakaways: $s \simeq 0.394$, $s \simeq -1.565$ — Locus doesn't exist ✗; real breakaway: $s \simeq -6.079$ — Locus exists ✓

S-Plane: **Locus:** **Exists:** $-3 < s < -9$, $-1 < s < -0.5$

Doesn't exist: $s < -9$, $-3 < s < -1$, $s > -0.5$